



ST BENEDICT'S

SUBJECT	PHYSICAL SCIENCE: CHEMISTRY
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mrs Govender
DATE	31 July 2024
MARKS	140
DURATION	2,2 hours

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LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 20 pages and a separate data sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Additional space is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	Total
Mark									
Moderated Mark									
Question Total	20	34	10	14	9	14	17	22	140

Comment:

QUESTION 1**14 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	C	D
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Here the option C has been marked as an example.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D
1.9	A	B	C	D
1.10	A	B	C	D

1.1 The correct chemical formula for ammonium dichromate is:

- A. $\text{NH}_4\text{Cr}_2\text{O}_7$
- B. $(\text{NH}_4)_3\text{Cr}_2\text{O}_7$
- C. $\text{NH}_4(\text{Cr}_2\text{O}_7)_2$
- D. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$

1.2 Which of the following will represent the largest number of moles of gas at **STP**?

- A. 10 cm^3 of ammonia (NH_3)
- B. 15 cm^3 of methane (CH_4)
- C. 20 cm^3 of nitrogen (N_2)
- D. 25 cm^3 of hydrogen (H_2)

1.3 Which substance will conduct electricity **ONLY** when molten?

- A. graphite
- B. gold
- C. iodine
- D. iron(II)sulphate

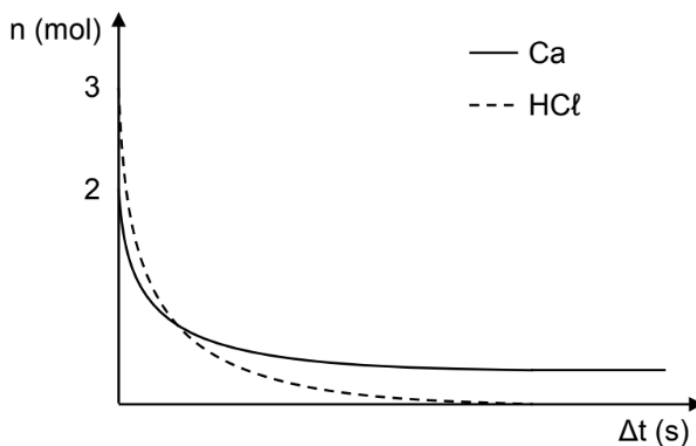
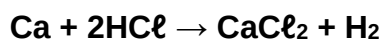
1.4 How many hydrogen atoms are present in 8,5 g of NH_3 ?

- A. $3,01 \times 10^{23}$
- B. $6,02 \times 10^{23}$
- C. $9,03 \times 10^{23}$
- D. $2,40 \times 10^{24}$

1.5 Which of the following has covalent intramolecular bonds that are the most polar?

- A. Na_2S
- B. SO_2
- C. SF_2
- D. SeS_2

1.6 Consider the balanced chemical equation below:



Which one of the following correctly identifies the limiting reagent and the amount of CaCl_2 produced?

	Limiting reagent	Amount of CaCl_2 produced (mol)
A.	Ca	1
B.	Ca	2
C.	HCl	1,5
D.	HCl	3

1.7 Which of the following is not an ionic compound?

- A. HF
- B. LiCl
- C. BeCl_2
- D. NH_4Cl

- 1.8 Which of the following is a non-polar molecule with polar covalent bonds?
- A. NH_3
 - B. CO
 - C. BF_3
 - D. OF_2
- 1.9 Which one of the following statements regarding activation energy is not correct?
- A. Activation energy is temperature dependent.
 - B. Activation energy is not influenced by temperature.
 - C. Different reactions have different values for activation energy.
 - D. A catalyst lowers the activation energy required by orientating molecules more favourably.
- 1.10 CO_2 has a boiling point of -78°C , whereas Cl_2 has a boiling point of -34°C . Which statement best explains the difference in boiling points:
- A. CO_2 forms London forces between the molecules while Cl_2 forms dipole-dipole forces which are stronger and use more energy to break.
 - B. CO_2 has instantaneous dipoles and Cl_2 induced dipoles forming between their molecules. Different strengths of forces thus form.
 - C. Cl_2 is not a dipole molecule, but CO_2 is and forms stronger bonds that must be broken.
 - D. Both are non-dipole molecules with London forces between the molecules but Cl_2 is forming more forces at the same time.

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QUESTION 2 CHEMICAL BONDING**34 MARKS**

The table below shows the melting and boiling points of compounds. Use the table below to answer the questions that follow.

Compounds	Melting point (°C)	Boiling point (°C)
Copper chloride	620,0	993,0
Water	0,0	100,0
Ammonia	-7,73	-3,34

2.1 Define the term *polar covalent bond*. (2)

2.2 Determine the bond polarity of ammonia. (3)

2.3 Draw the Lewis diagram of ammonia. (3)

2.4 Determine the molecule polarity of ammonia. (5)

2.5 Indicate the molecule polarity on your Lewis diagram. (2)

2.6 Name the type of intermolecular forces that exists between **ammonia** molecules. (1)

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2.7 With reference to intermolecular forces, compare the boiling points of water and ammonia and fully explain this difference. (5)

2.8 Define the term *ionic bonding*. (2)

2.9 Use Lewis Dot notation to show the formation of copper chloride. (4)

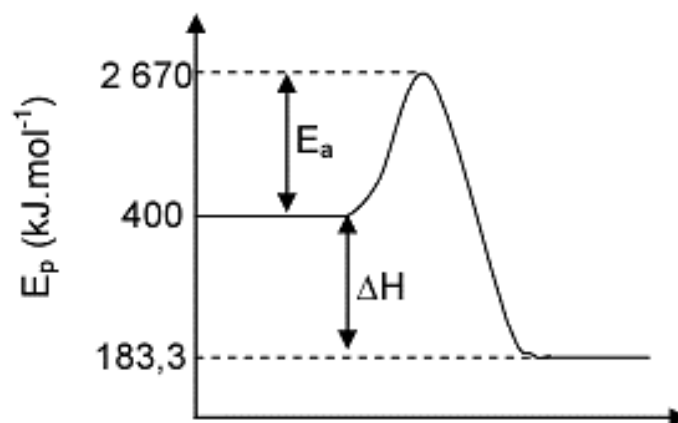
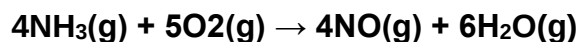
2.10 Write the half reactions for copper and chlorine. Label the reaction as oxidation or reduction. (4)

Copper:
Chlorine:

2.11 Which compound from the table is a good conductor of electricity in a liquid phase? Explain your answer. (3)

QUESTION 3 REACTION RATES**10 MARKS**

The following reaction between ammonia and oxygen takes place in a closed system at constant pressure and temperature:



3.1 Define the term *activation energy*. (2)

3.2 Does this energy profile diagram represent **exothermic** or **endothermic** reaction? Circle the correct answer. (1)

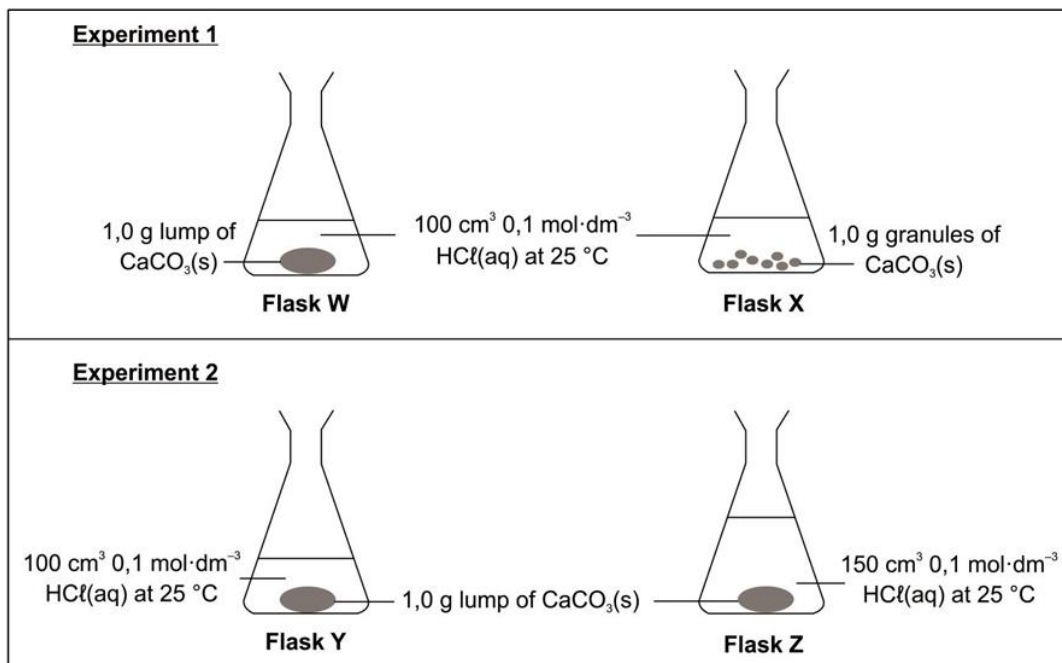
3.3 Explain your answer in 3.2. (2)

3.4 Calculate the heat of reaction. (3)

3.5 Redraw the graph and indicate with a dotted line the effect of a catalyst on the activation energy. (2)

QUESTION 4 REACTION RATES**14 MARKS**

Two experiments were conducted to investigate the rate of reaction between EXCESS calcium carbonate and dilute hydrochloric acid. The balanced chemical equation for the reaction is given below.



- 4.1 Suggest a suitable method of measuring the rates of the reactions in these experiments. Additional apparatus besides that shown in the diagrams may be used. State what apparatus is required and what reading(s) would need to be taken. (3)

- 4.2 In Experiment 1, in which flask, **W** or **X**, will the reaction rate be faster? Circle the correct answer. (1)
- 4.3 Explain your answer in 4.2. (3)

4.4 Calculate the volume of $\text{CO}_2(\text{g})$ produced at STP in flask **W**. (4)

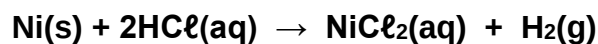
4.5 Consider the reaction conditions given for Experiment 2.

4.5.1 How will the rate of reaction in flask **Z** compare to that in flask **Y**? Choose from GREATER THAN **Y**, LESS THAN **Y** or EQUAL TO **Y**. (1)

4.5.2 It is observed that the total volume of gas produced in flask **Z** is greater than that produced in flask **Y**. Account for this observation. (2)

QUESTION 5 REACTION RATES**9 MARKS**

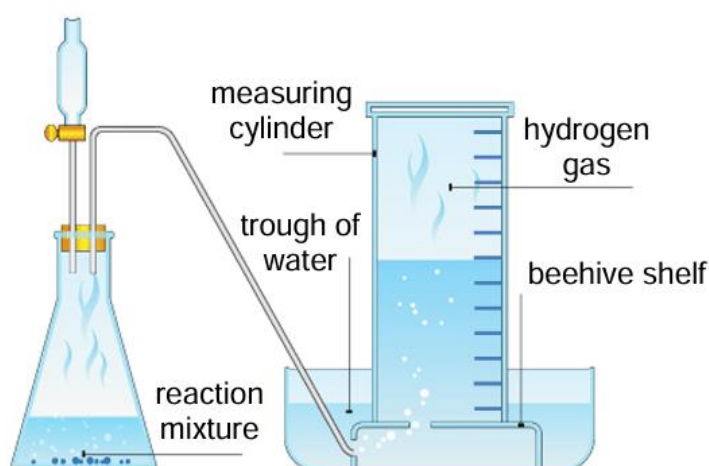
The reaction of nickel metal with hydrochloric acid is represented below.



- 5.1 Which TWO conditions have to be met for an effective collision to occur between colliding reactant particles? (2)

- 5.2 Use the collision theory to explain how an increase in the concentration of HCl would affect the rate of this reaction. (3)

In an experiment, Kristen collects the hydrogen gas. The setup is shown below.



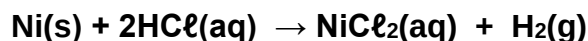
Kristen measures the volume of gas produced over time in two experiments, Experiment 1 and Experiment 2.

In both experiments, Kristen uses the same mass and state of nickel; the acid is the limiting reagent and the nickel is completely covered by the acid solution.

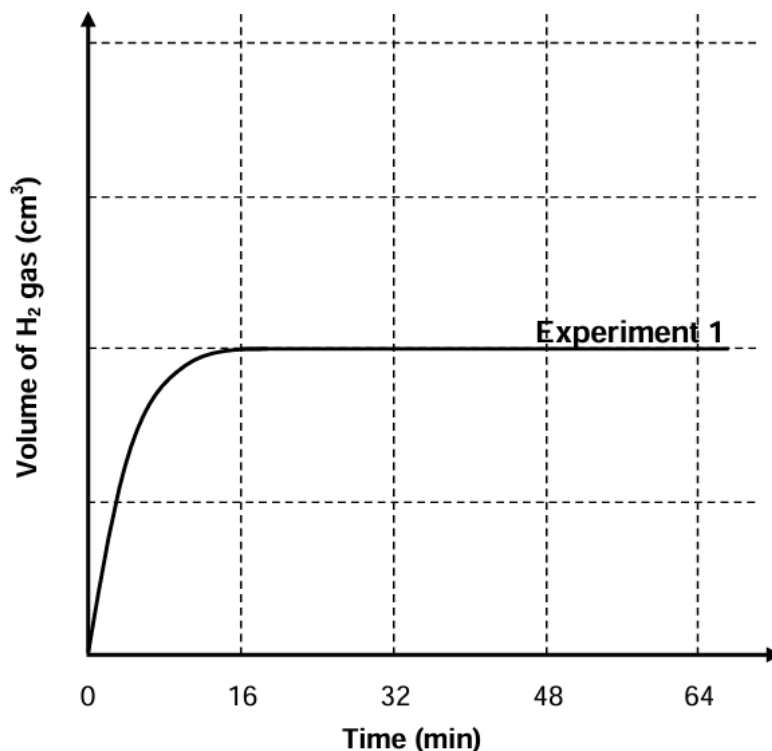
The reaction conditions for each experiment are tabulated below.

Experiment	Volume of HCl (dm ³)	Concentration of HCl (mol·dm ⁻³)	Temperature (°C)
1	0,4	0,04	50
2	0,6	0,04	40

The balanced chemical equation for the reaction of nickel with hydrochloric acid is repeated here:



Kristen plots a graph of volume of hydrogen gas produced versus time for Experiment 1, which is shown below.

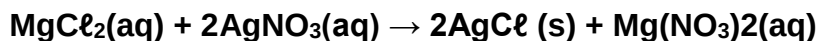


5.3 Describe the method used to collect the gas. (2)

5.4 On these same axes, draw the graph for Experiment 2. Assume this reaction goes to completion within the 64 minutes shown. (2)

QUESTION 6 STOICHIOMETRY**14 MARKS**

Two solutions, one being silver nitrate (AgNO_3) and the other, magnesium chloride (MgCl_2) are mixed together in a glass container. Solid silver chloride (AgCl) forms at the bottom of the container as seen in the following balanced reaction that took place:



55 cm³ of the magnesium chloride, with a concentration of 0,25 mol·dm⁻³ was mixed with 70 cm³ of silver nitrate solution with a concentration of 0,15 mol·dm⁻³.

6.1 Define the term *molar mass*. (2)

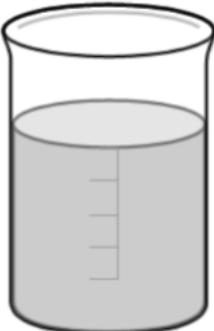
6.2 Prove that the silver nitrate is the limiting reactant. (6)

6.3 Hence, calculate the mass of silver chloride that can formed during this reaction. (3)

6.4 Calculate the percentage yield of silver chloride if 1,49 g of silver chloride was in fact recovered. (3)

QUESTION 7 STOICHIOMETRY**17 MARKS**

Consider the following two aqueous solutions:

Solution A $c = 0,2 \text{ mol} \cdot \text{dm}^{-3}$ ammonium hydroxide $V = 0,25 \text{ dm}^3$			Solution B $c = 0,3 \text{ mol} \cdot \text{dm}^{-3}$ ammonium carbonate $V = 0,5 \text{ dm}^3$
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7.1 Give a reason why the above solutions are standard solutions. (2)

7.2 Define the term *solvent*. (2)

7.3 Identify the solvent used in the above solutions. (1)

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7.4 Give the method for making a standard solution using four key bullet points. (4)

The balanced dissociation reaction for ammonium hydroxide is given as:



- 7.5 Write balanced chemical equation showing the dissociation of ammonium carbonate. (3)

- 7.6 Solution A is now mixed with Solution B in a new bigger beaker. Calculate the new concentration of the ammonium ions in the solution. (5)

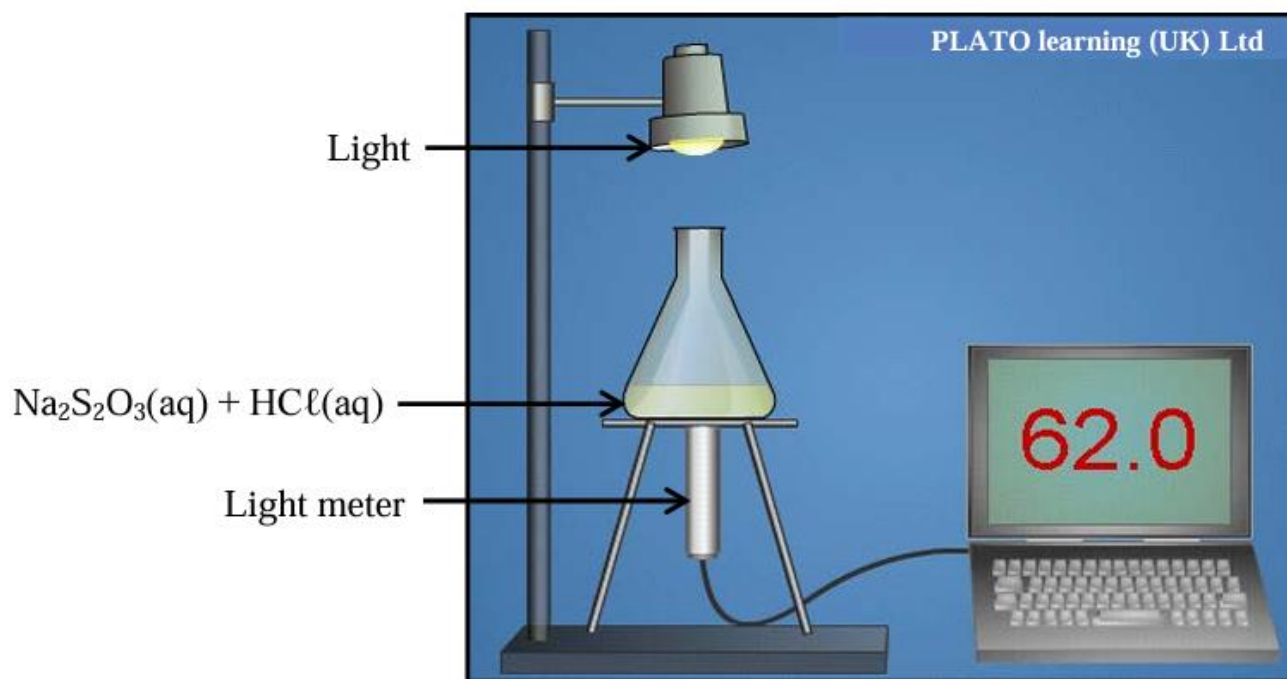
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QUESTION 8 SCIENTIFIC SKILLS**22 MARKS**

Consider the reaction of sodium thiosulphate solution with dilute hydrochloric acid as given by the balanced chemical equation below.



The rate of reaction between the sodium thiosulphate solution and hydrochloric acid can be monitored using a light meter, which measures the decrease in the intensity of light passing through the solution as the reaction proceeds. The apparatus is set up as shown in the diagram below.



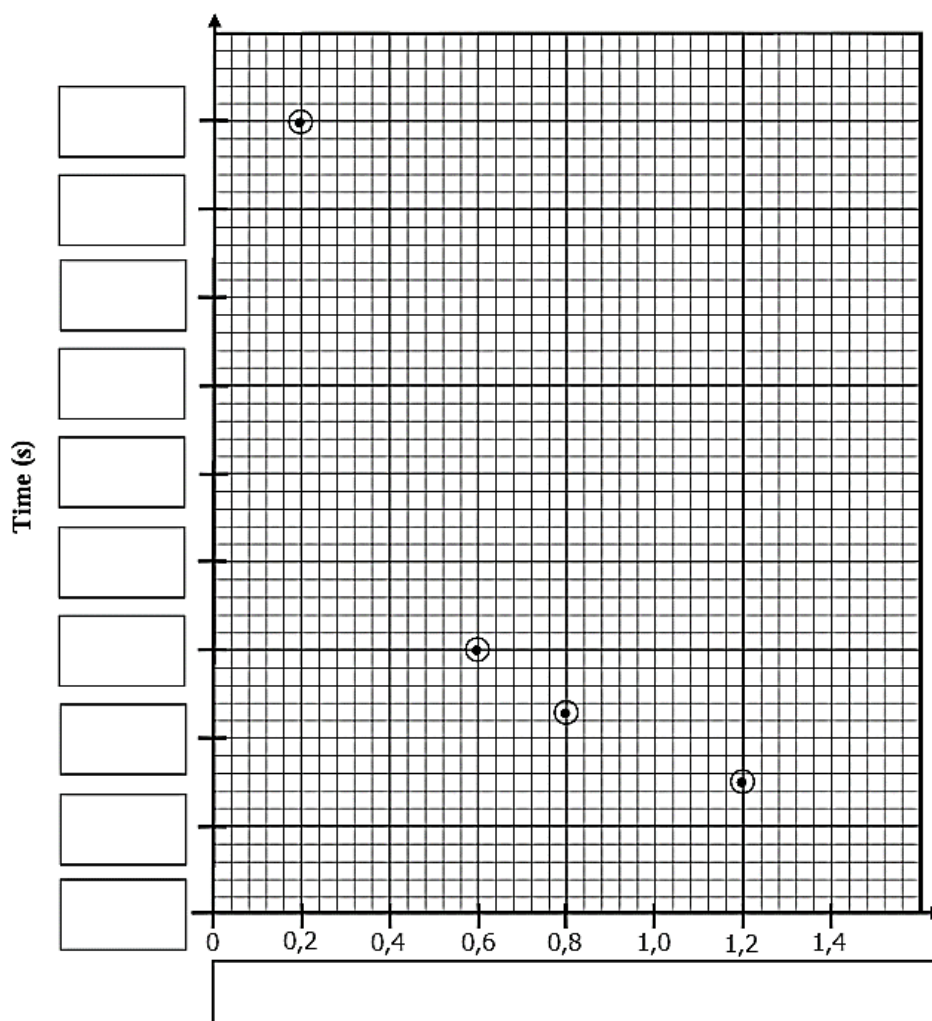
Luke uses the apparatus shown in the diagram to conduct a series of experiments to determine the relationship between the concentration of the sodium thiosulphate solution and the time taken for the light meter to show a reading of 62 lumens. (Lumens = unit for light intensity.)

The results of the experiment are shown below.

	Concentration of $\text{Na}_2\text{S}_2\text{O}_3$ ($\text{mol} \cdot \text{dm}^{-3}$)	Time (s)
Experiment 1	0,2	180
Experiment 2	0,4	90
Experiment 3	0,6	60
Experiment 4	0,8	45
Experiment 5	1,0	36
Experiment 6	1,2	30

- 8.1 Why will there be a decrease in the intensity of the light passing through the solution as the reaction proceeds? (2)

Luke plotted a graph to show the relationship between the concentration of the $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and the time taken for the light meter to show a reading of 62 lumens. The incomplete graph is provided below.



- 8.2 Write down a suitable label for the x-axis in the empty block provided. (2)
- 8.3 Determine the scale used on the y-axis and fill in the numbers in the empty blocks provided. (1)
- 8.4 Plot the points for experiments 2 and 5. (2)
- 8.5 Draw a best-fit line through the points. (2)
- 8.6 State a reason why Luke plotted time on the y-axis rather than on the x-axis. (1)

8.7 Write the independent variable for this experiment. (2)

8.8 State the relationship in words for this experiment. (2)

8.9 Write the conclusion for this experiment. (2)

8.10 Comment on the reliability for this experiment. (3)

8.11 Comment on the accuracy of experiment 5 if it actually takes 41s for the light intensity to decrease. (3)

ADDITIONAL SPACE

[illegible]
